DUAL-SPACE DISENTANGLEMENT AND REVERSION FOR ROBUST BACKDOOR DEFENSE

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ABSTRACT

This paper addresses the challenge of backdoor attacks in text-to-image diffusion models, where maliciously embedded triggers can cause unauthorized content generation, including copyrighted imagery. We focus on overcoming a key limitation of existing approaches such as SilentBadDiffusion, which rely on pixel-level or semantic fragment insertion but offer insufficient detection and mitigation at inference. We propose Dual-Space Disentanglement and Reversion (DSDR), a new method that operates jointly in the semantic (image) domain and the latent-noise space. DSDR encodes small but critical fragments of copyrighted data in noise signatures, then monitors them during inference. It can detect subtle triggers by measuring the KL divergence from benign noise distributions and revert suspicious noise patterns. We verify the method's effectiveness through controlled backdoor embedding, quality assessment, and an ablation study. Experimental results show that DSDR not only reduces successful backdoor activations but also preserves image fidelity, helping protect diffusion models from unauthorized modifications while maintaining broad applicability in text-to-image generation.

1 INTRODUCTION

Generative artificial intelligence has experienced remarkable growth in recent years, especially in text-to-image diffusion models (DMs). These models are now widely available through online services and commercial platforms, raising concerns about intellectual property rights, privacy leaks, and malicious manipulations of model behaviors. Among such concerns, backdoor attacks stand out: attackers subtly embed triggers into the training pipeline so that, when targeted prompts are used during inference, the model outputs content that violates certain protections or reveals hidden information.

Recent research on SilentBadDiffusion illustrates the gravity of such backdoor threats. By carefully injecting pieces of copyrighted images in training data, this approach can link references to copyrighted material with benign text prompts, thus enabling the re-generation of protected content. However, these methods depend strongly on controlling either fine-tuning stages or injecting consistent pixel modifications, and they often assume that disentangling attacks relies on only one or two metrics (like a similarity measure). In practice, advanced diffusion models have more complex internal structures that can circumvent simplistic detection methods. The phenomenon that "the more sophisticated the DMs are, the easier the success of the attack becomes" underscores how stealthy attackers take advantage of the model's large capacity.

In this work, we propose a method that builds upon the strengths of prior work, specifically leveraging the idea of segmenting protected content and embedding it subtly. Yet we expand to dual-space detection by introducing the concept of noise-space triggers and a continuous reversion process. This new framework, called Dual-Space Disentanglement and Reversion (DSDR), aims to address core vulnerabilities of existing solutions.

- **Dual-space approach:** We formalize a dual-space approach for identifying, suppressing, or removing backdoor triggers, combining semantic-level disentanglement with robust latent-noise analysis.

- **Noise-based trigger reversion:** We design a noise-based trigger reversion mechanism that continually monitors the KL divergence between a model's latent states and a reference "benign" distribution, allowing for timely reversion of suspicious noise patterns.
- **Comprehensive experiments:** We present a comprehensive set of experiments to demonstrate the reliability of DSDR. By comparing to baseline approaches, including SilentBadDiffusion-style methods, we showcase improved detection rates and negligible impact on normal image generation.
- **Extension outline:** We provide an outline for extending DSDR to more complex attacks, including scenarios where images cannot be neatly decomposed or where multiple types of copyrighted fragments appear.

The remainder of the paper reviews related work, details relevant background concepts, describes the DSDR approach in depth, followed by our experimental setups and implementation specifics. We conclude with a discussion of the key findings, limitations, and directions for future research.

2 Related Work

Backdoor attacks against machine learning models have been widely investigated across various domains, such as image classification, speech recognition, and language modeling. Typical strategies rely on inserting triggers into training data so that the network learns to associate specific cues with adversarial outputs. In diffusion-based generative models, backdoors represent a unique challenge: the generative process inherently involves sampling from a noise space, and the prompt-based conditioning can be exploited for surreptitious triggers.

Prior works on diffusion backdoors often adopt pixel-level manipulations, where an attacker modifies training images or textual embeddings to embed malicious signals. These triggers can be a small visible patch, an imperceptible noise pattern, or even a distribution-shifted style that the model memorizes. The method known as SilentBadDiffusion, for instance, hides pieces of copyrighted images across multiple training samples, linking them to certain text prompts so that the model eventually regurgitates copyrighted material at inference time.

Attacks that embed triggers in generative models differ from classification backdoors in two main ways. First, there is a higher risk of intellectual property or privacy infringement: generative models can recreate sensitive data once triggered, rather than simply misclassifying an input. Second, detecting these triggers generally requires a deeper understanding of how the generative process unfolds in noise space. While some defenses focus on scanning model outputs using perceptual similarity detectors, such approaches can fail if the copyrighted fragments are recombined in subtle ways.

Backdoor detection strategies typically rely on data-level examination, model-level analysis, or inference-time monitoring. Data-level methods attempt to locate suspicious shifts in the training set, but may be ineffective when triggers are dispersed. Model-level analysis can examine latent representation distributions for anomalies, while inference-time defenses often track output statistics to apply post-hoc corrections when anomalies are detected.

Our proposed DSDR method builds on these lines of research but integrates semantic and noise-level detection more comprehensively, ensuring that both the visual fragment and its latent noise signature are jointly considered in defending against backdoor attacks.

3 Background

Diffusion models generate high-quality images based on iterative denoising processes guided by textual or other conditional inputs. The typical pipeline starts with a random noise vector, which is gradually refined until it aligns with the distribution learned from the training data. In text-to-image diffusion, user-provided prompts steer the generation process by providing semantic structure through transformer-based or CNN-based encoders.

Due to the reliance on large datasets, these models can inadvertently memorize or reproduce undesirable patterns, making them vulnerable to issues such as copyright infringement or privacy leaks. SilentBadDiffusion exemplifies this risk by dissecting copyrighted images into segments and scattering them across multiple images and text references. Through training on a poisoned dataset, the model learns to associate benign prompts with protected visual patterns.

Advanced diffusion models utilize highly flexible latent representations that can capture minute details. Consequently, even dispersed fragments may be reassembled when certain instructions are encountered. The challenge of selectively removing backdoor triggers is amplified by the multi-step generative pipeline, where a small latent perturbation can lead to significant output changes.

Formally, the diffusion process can be seen as sampling x from a distribution p(x) and iteratively refining x into x by removing noise, guided by learned parameters. For a backdoor to succeed, an attacker aims to manipulate the training process via an adversarial distribution p*(x) such that specific factors reappear when a corresponding condition c is provided. DSDR addresses this challenge by monitoring hidden noise embeddings and comparing them to a benign reference distribution, thus isolating suspicious patterns through KL divergence measures.

4 METHOD

We introduce Dual-Space Disentanglement and Reversion (DSDR) as a unified mechanism to detect and neutralize silent backdoor triggers in text-to-image diffusion models. Our approach consists of two primary phases—embedding and reversion—and leverages formalism from the problem setting described in the background.

4.1 OVERALL FRAMEWORK

DSDR builds on the concept of semantically decomposing a protected image into salient components, similar to the SilentBadDiffusion approach. We extend this idea by linking each visual fragment to a corresponding noise signature. At training, if the model learns to associate a protected component with a particular code in noise space, it can later regenerate it only when that noise code is reencountered.

4.2 DUAL-SPACE DISENTANGLEMENT

The disentanglement process comprises two parts:

- **Semantic Decomposition:** An automated segmentation toolkit extracts key elements of an image (such as outlines, texture details, and background). Each fragment is treated as a separate entity, particularly for copyrighted materials.
- **Latent Anchoring:** Each extracted fragment is associated with a unique signature in the latent-noise space. An encoder fuses the fragment's features into a noise vector, which is distributed broadly across the latent space. The semantic reconstruction of an image thus depends on the simultaneous presence of the visual fragment and its corresponding noise signature.

4.3 TRIGGER REVERSION AND MONITORING

During inference, DSDR continuously monitors the generation process. The key steps include:

- **Reverse-sampling:** A batch of images is generated from random Gaussian noise. Intermediate noise vectors are compared against a benign reference distribution.
- **KL-Divergence Guided Correction:** For each intermediate noise vector, the KL divergence from the benign distribution is computed. If it exceeds a predefined threshold, a reversion step is triggered to push the noise vector closer to the safe region.
- **Continuous Monitoring:** The system updates an internal record of potential trigger patterns at every time step. Consistent reappearance of a pattern across iterations or prompts increases confidence that it is malicious, prompting further reversion.

The following pseudocode outlines the noise reversion process:

Algorithm 1 Noise Reversion Process

```
Initialize benign reference distribution R
for each intermediate noise vector x_t do
    Compute KL divergence: D_KL ← KL(x_t || R)
    if D_KL > θ then
        Adjust x_t ← x_t − α(x_t − R)
    end if
end for
```

4.4 Advantages

DSDR offers several benefits:

- **Flexibility:** The method works effectively for both decomposable images and those where only latent noise signatures can be reliably extracted.
- **Unified Pipeline:** By integrating detection and reversion into a single process, the approach simplifies the overall defense mechanism.
- **Adaptability:** The KL-divergence threshold and number of reversion iterations can be tuned to balance responsiveness with the preservation of normal image generation quality.

4.5 Potential Extensions

Future work may involve dynamic thresholding based on semantic variability, integrating multi-modal triggers (e.g., text, image, audio), and watermarking non-reverted outputs to signal model integrity.

5 Experimental Setup

We evaluate DSDR through three distinct experiments comparing its performance against baseline models, including a standard semantic segmentation approach and a SilentBadDiffusion-inspired pipeline. The experiments are implemented in Python using frameworks such as PyTorch, torchvision, and scikit-image.

5.1 Controlled Backdoor Embedding and Inference

Objective: Validate DSDR's ability to detect and revert stealthy triggers.

Data Generation: A synthetic dataset is constructed from public sources (e.g., CIFAR-10 or a subset of ImageNet), with approximately 10% of the images embedded with latent triggers that simulate backdoor insertion via small noise patterns.

Training: Two models are trained: one baseline model without any reversion step and a DSDR-augmented model that monitors KL divergence and reverts suspicious noise. Both models are optimized with a reconstruction-based loss mimicking text-to-image generation objectives.

Inference: Reverse sampling from Gaussian noise is performed, and anomalies are tracked via computed KL divergence. When divergence exceeds a chosen threshold, the DSDR mechanism reverts the latent vectors toward a benign distribution. Metrics include trigger detection rate (true positives) and false negatives.

5.2 Quality and Consistency Evaluation of Image Outputs

Objective: Assess the impact of DSDR's noise reversion on overall image quality.

Setup: A set of prompts is used—some benign and others intended to activate triggers. Both the baseline and the DSDR model generate images for these prompts.

Evaluation Metrics: Standard image quality measures such as FID (Fréchet Inception Distance) and SSIM (Structural Similarity Index) are computed to assess differences between generated and real distributions, as well as pixel-level fidelity.

Procedure: Comparisons are made between the frequency of triggered outputs from the baseline and the reverted outputs from DSDR. Any degradation in benign image quality is monitored through FID and SSIM values.

5.3 Ablation Study on KL-Divergence Threshold and Reversion Steps

Objective: Investigate the trade-off between detection aggressiveness and image quality preservation.

Setup: A grid of hyperparameters is defined with varying KL threshold values (e.g., 0.05, 0.1, 0.15, 0.2) and different numbers of reversion iterations (e.g., 1, 3, 5).

Method: Each configuration is evaluated based on detection rates, reconstruction loss, and inference times. Metrics are recorded to identify the optimal balance between robust trigger suppression and minimal impact on benign content.

Implementation Details: All experiments are encapsulated within a single script (e.g., dsdr$_{e}xperiments.py)thathandlesdatasetcreation, modeltraining, inferencemonitoring, andmetricloggingusingTen$

6 Results

The experimental results demonstrate the strength of the DSDR approach in terms of trigger detection, image quality, and parameter sensitivity.

6.1 Trigger Detection and Reversion Analysis

Experiment 1 indicates that DSDR-augmented models detect and revert a significantly higher proportion of embedded triggers compared to the baseline. Under typical poison ratios (around 10% in the synthetic dataset), the baseline model frequently activates malicious patterns, whereas DSDR, with its KL-divergence monitor and iterative reversion, substantially reduces successful attacks.

6.2 Image Fidelity and Consistency

From Experiment 2, both models generate images of comparable quality on benign prompts, as measured by FID and SSIM. However, when backdoor triggers are present, DSDR produces images with fewer suspicious artifacts and distributions more closely aligned with clean data.

../images/training_loss.pdf

Figure 1: Training Loss Over Time

../images/quality_metrics.pdf

Figure 2: Quality Metrics Comparison between Baseline and DSDR

../images/ablation_kl.pdf

Figure 3: Ablation Study Results on KL-Divergence Threshold and Reversion Steps

6.3 Extended Observations and Limitations

While the approach assumes that KL divergence is an effective measure for distinguishing benign from malicious patterns, future work may address adaptively shifting triggers or non-decomposable image content. The experiments suggest that integrating semantic and noise space detection into a unified pipeline offers enhanced resilience against backdoor strategies, with flexible hyperparameter tuning to suit varying threat scenarios.

7 Conclusions and Future Work

Dual-Space Disentanglement and Reversion (DSDR) has been introduced as a systematic approach to mitigate stealthy backdoor attacks in text-to-image diffusion models. By jointly addressing semantic fragmentation and noise-based triggers, DSDR swiftly detects and reverts malicious embeddings without compromising normal image generation.

Experimental results confirm that DSDR outperforms simpler baseline models in suppressing triggers, even when the poisoning fraction is low, and maintains high image fidelity on benign content. Future work will explore extending DSDR to handle non-decomposable images and multi-modal backdoor strategies, as well as refining adaptive thresholds and iterative reversion protocols. Overall, DSDR represents a significant step toward safer diffusion pipelines capable of protecting intellectual property and ensuring the integrity of generative outputs.

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